

Accuracy of Rapid On-Site Evaluation in Robotic-Assisted Bronchoscopy Fine Needle Aspirations of Lung Nodules

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Background: Shape-sensing robotic-assisted bronchoscopy (ssRAB) is an emerging technique for sampling lung nodules. When evaluating the value of pairing ssRAB with rapid on-site evaluation (ROSE), other studies have only looked at diagnostic yield. This study aims to assess the diagnostic concordance of ROSE and histopathologic biopsies in ssRAB sampling.

Methods: We analyzed lung nodules sampled by ssRAB fine needle aspiration (FNA) and had ROSE. The following parameters were recorded: preliminary and final cytology diagnoses, concurrent histopathology biopsy diagnosis, tool type, number of passes for ROSE, lesion size, and location. A χ^2 test was performed for categorical variables, and one-way analysis of variance was used to compare means for continuous variables. A P -value < 0.05 was considered statistically significant.

Results: A total of 249 FNA cytology specimens from 234 patients were obtained, with 202 concurrent histopathologic biopsies. The lesion size, location, and proceduralist experience did not affect cytology diagnosis; however, the diagnostic yield improved with increasing lesion size. Diagnostic yield plateaued after 3 passes ($P = 0.01$). Forceps biopsies had a lower diagnostic yield (74%) than cryoprobe biopsies (87%), and forceps were more likely to yield a nondiagnostic sample than cryobiopsies ($P = 0.04$). A positive ROSE preliminary diagnosis for malignancy correlated highly with a positive histopathology biopsy (86% correlation).

Conclusion: A positive result for malignancy on ROSE correlated highly with a positive histopathology biopsy. Three fine needle passes were sufficient for diagnostic results, with a plateau in subsequent biopsies. Histopathologic tissue diagnosis was superior with the cryoprobe than with tissue obtained using conventional forceps.

Key Words: lung nodule, cytology, robotic bronchoscopy

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Several modalities of guided bronchoscopy for sampling of lung nodules have been the mainstay approach for minimally invasive tissue acquisition for diagnosis and ancillary testing. However, the diagnostic yield has remained at about 70%.¹ The recently published VERITAS study, a randomized multicenter clinical trial compared both electromagnetic navigational bronchoscopy (ENB) and CT-guided biopsies for tissue sampling of lung nodules, showed non inferiority of the diagnostic yield of bronchoscopy to the CT-guided approach (79.0% and 73.6%, respectively), with a better safety profile favoring the bronchoscopy group (pneumothorax rate 3.3%) when compared with the CT-guided group (pneumothorax rate 28.3%).² Shape-sensing robotic-assisted bronchoscopy (ssRAB) is an emerging technique. A preprocedural CT scan is used to create a 3-dimensional (3D) map of the patient's airways. The target lesion is marked, and a virtual pathway to it is created. During the procedure, after automatic registration, the proceduralist navigates to the target lesion. The RELIANT multicenter trial has shown similar diagnostic yields and safety profiles between ssRAB and ENB, when both have intraprocedural advanced imaging—cone beam CT (CBCT) and digital tomosynthesis—in the robotic and ENB group, respectively.³

Prior published studies have only examined the final cytology diagnosis to evaluate ssRAB's diagnostic yield with lung nodule sampling.^{3–8} The value of rapid on-site evaluation (ROSE) for airway procedures is in (a) preventing unnecessary additional biopsies by targeting the highest-stage thoracic lymph nodes and (b) ensuring sufficient diagnostic tissue in heterogeneous and difficult-to-biopsy pulmonary lesions.^{9–11} The diagnostic yield for a combination of ssRAB fine needle aspiration (FNA) and a histopathologic biopsy obtained with a cryoprobe can range from 74% to 80%.^{6–8} In addition, it has been reported that ssRAB FNAs have a sensitivity for cancer of 84%, and for definitive malignancy is 88%.^{12,13} Importantly, a meta-analysis of bronchoscopies using radial endobronchial ultrasound (rEBUS) showed that the only significant factor increasing sensitivity for malignancy in biopsy was the use of ROSE.¹⁴ As ssRAB has a higher likelihood of navigating into smaller, peripheral lesions, and the use of CBCT is becoming standard among operators, evaluating the necessity of ROSE has become a topic of interest, particularly regarding the reduction of procedural time and radiation exposure to team members.¹⁵

This single-institution, retrospective study aims to evaluate the diagnostic concordance between ROSE FNA, final cytology, and histopathology lung nodule sampling using ssRAB, and to determine the optimal number of

passes for ROSE evaluation. Secondary aims were to identify characteristics related to the lung nodule and ssRAB procedure that might affect the diagnostic yield, especially for small peripheral lesions.

MATERIALS AND METHODS

Intervention

Patients who were assigned to undergo ssRAB for lung nodule biopsy received general anesthesia for the procedure. Standard ventilator settings included tidal volumes of 6 to 8 mL/kg, starting positive end-expiratory pressure (PEEP) of 10 to 15 cm H₂O, and the lowest tolerated FiO₂ to maintain oxygen saturations >92%. After initial airway inspection, the ssRAB platform (ION; Intuitive Surgical, Inc., Sunnyvale, CA) was docked to the endotracheal tube, and the catheter was advanced into the area of interest. Prior CT planning was performed using software provided by the system, based on a CT scan obtained within 3 months before the procedure. After target navigation, rEBUS was inserted to confirm the position of the nodule in relation to the catheter, followed by FNA, and other biopsies at the discretion of the operator—cryoprobe 1.1mm (ERBE) or/single-use jaw forceps 4 (Boston Scientific). All sampling was guided by 2D fluoroscopy, and no use of CBCT or digital tomosynthesis was made during this study. For all cases, the ROSE team was available in the procedural room.

Data Collection

Cases were collected from lung ssRAB FNAs that occurred from November 1, 2022, to March 25, 2024. This study was approved by the Michigan Medicine Institutional Review Board (HUM00252213). The following data points were collected for all cases: age, sex, biopsy location (lung lobe, lung segment, and peripheral/central), reported lesion size during the procedure, preliminary cytology, and final cytology diagnosis.

For the FNAs, the data collected included the number of air-dried slides made, the number of additional passes for cell block or other ancillary studies, and the total number of passes reported. If a histopathologic biopsy was concurrently collected, the histopathologic diagnosis and the method of collection (using forceps and/or a cryoprobe) were recorded. If a patient did not undergo a concurrent diagnostic histopathologic biopsy, it was recorded whether there was later histopathologic confirmation (resection or repeat histopathologic biopsy) or if there was only clinical follow-up.

For the procedure, data collected included proceduralist information, rEBUS data, whether the procedure was a repeat ssRAB FNA, and any procedural complications. Lastly, if a concurrent histopathologic biopsy was not taken, the reason was noted in the procedure note.

Diagnosis Categorization

The preliminary and final cytology diagnoses were subsequently categorized according to the diagnostic criteria outlined in the 1st edition of the WHO Reporting System for Lung Cytopathology:¹⁶ nondiagnostic, negative, atypical, suspicious, or positive. The definition of a diagnostic histopathologic biopsy was based on the American Thoracic Society/American College of Chest Physicians Research Statement; specifically, a malignant or benign diagnosis by a pathologist was established and

sufficient to inform patient care.¹⁷ For histopathologic biopsies, the diagnoses were categorized by the same 5 cytology categories (nondiagnostic, negative, etc). Histopathologic biopsies were considered positive or suspicious if they were positive for neoplasm, including carcinomas (adenocarcinoma, squamous cell, and small cell), carcinoid tumors, or lymphomas. Biopsies were considered nondiagnostic if the diagnosis mentioned insufficient sampling of alveolar tissue (ie, scant tissue or bronchial wall only) and atypical if atypical cells were present. Biopsies that were called negative were those that were negative for malignancy, which included diagnoses showing adequate benign alveolar lung tissue without diagnostic abnormalities or non-neoplastic processes, such as organizing pneumonia or granulomatous inflammation.

Outcomes

For diagnostic outcomes, tissue samples obtained by bronchoscopy (using forceps or a cryoprobe) or histopathologic resection (such as wedge resection or lobectomy) were confirmed through histopathology. If a histopathologic specimen was not obtained, a nodule was considered at least suspicious for malignancy by multidisciplinary tumor board consensus, or the patient died shortly after, and it was presumed to be due to malignant disease. Similarly, for those lesions without a histopathologic sample and nondiagnostic FNA, a nodule was considered clinically benign if the lesion was attributed to a benign cause (such as infection or inflammation), no oncological/surgical treatment was recommended by the tumor board, and the nodule size decreased/showed no change on subsequent imaging with a 1-year follow-up.

Statistical Analysis

Statistical analysis was conducted to see the correlation between obtaining diagnostic cytology tissue and the number of passes, lesion size, lesion location (lobe, segment, and peripheral/central), and whether the lesion was identified on rEBUS. Peripheral lesions were defined as located in the outer third of the lung parenchyma as observed in the CT chest by a blinded reviewer. If the lesion was located within the inner two-thirds of the lung, it was labeled as central. Two different groups were created for statistical analysis: the 5 diagnostic categories as standalone and the 3 combined diagnostic categories. These combined categories were clinically nondiagnostic (atypical and nondiagnostic), clinically positive (suspicious and positive categories), and clinically negative (negative alone). Concordance was measured to see how often the preliminary cytology diagnosis agreed with the final cytology diagnosis and the histopathologic biopsy. Diagnoses were considered discordant if the diagnostic category was completely overturned (ie, from negative to positive or from positive to negative).

For categorical variables, we utilized the χ^2 test to determine the probability of independence between 2 categorical variables. We used one-way analysis of variance to compare the means among several groups for continuous variables. Comparisons with a *P*-value of <0.05 were considered statistically significant.

RESULTS

There were 249 cytology specimens from lung nodules identified from 234 unique patients, with 202 concurrent histopathology biopsies (forceps and cryoprobe biopsies) (Supplement A, Supplemental Digital Content 1, [2 | \[www.bronchology.com\]\(http://www.bronchology.com\)](http://</p></div><div data-bbox=)

links.lww.com/LBR/A394). Four proceduralists were involved in the study, with years of practice in advanced diagnostic bronchoscopy ranging from 2 to 22 years. All patients underwent an FNA biopsy, and 46% received a forceps biopsy (n = 94), 49% received a cryobiopsy (n = 98), and 5% had both a forceps and a cryobiopsy (n = 10). Six patients had a repeat ssRAB biopsy during the study period (2% of the cohort). The average size of the biopsied lesions was 1.8 cm, and each procedure required an average of 2.6 passes for ROSE adequacy. Additional demographic, lesion, and procedural information is detailed in Table 1, and the distribution of diagnoses is detailed in

Supplemental A, Supplemental Digital Content 1, <http://links.lww.com/LBR/A394>.

Nineteen patients had complications during ssRAB biopsy (8% of all patients). The most common complications were pneumothorax (n = 9) and nonmassive bleeding (n = 8). Of these complications, 12 involved peripheral nodules. There was no mortality during the study period.

Diagnostic Distribution Versus Nodule and Procedure Characteristics

The diagnostic yield for cytology specimens was 61%, and for histopathologic biopsy specimens for all lung nodules

TABLE 1. Demographic, Nodule, and Procedure Information

Demographics, N = 234 subjects, n (%)

Male/Female	Male: 109 (46.6) Female: 125 (53.4)
Average age (SD), y	70 ± 12
Nodule characteristics, n = 249 cytology specimens	
Average size (SD), cm	1.8 (1.21)
Lobe location	Left lower: 45 (18) Left upper: 58 (23) Right lower: 60 (24) Right middle: 16 (7) Right upper: 70 (28)
Segment location	B1: 34 (13) B1/2: 33 (12) B2: 20 (9) B3: 31 (8) B4: 18 (12) B5: 8 (3) B6: 40 (16) B7: 3 (1) B7/8: 5 (1) B8: 11 (4) B9: 21 (9) B10: 22 (9)
Atelectasis risk	Low: 165 (66) Medium: 20 (8) High: 62 (25) Undetermined: 2 (< 1)
The risk of atelectasis was based on the location of the nodule, as determined by the proceduralist. High-risk lesions were those located in the lower lobes, posterior, lateral, and superior segments in the outer third periphery.	
Peripheral lesion	Yes: 129 (52) No: 117 (47) Undetermined: 3 (1)
Procedure details	
Concurrent histopathologic biopsies by bronchoscopy	202 (79)
Types of biopsies, (% of 198)	Forceps: 94 (47) Cryoprobe: 98 (49) Forceps and cryoprobe: 10 (5.5)
Radial-EBUS (rEBUS) visualization, (% of 250)	Yes: 226 (91) No: 21 (8) Not indicated: 2 (1)
Proceduralist (#), # of cytology specimens, (% of 250)	#1: 42 (17) #2: 112 (45) #3: 65 (26) #4: 29 (12) #1 and #2: 1 (< 1)
Number of passes for ROSE, (% of 250)	1: 49 (20) 2: 80 (32) 3: 71 (28) 4: 30 (12) > 5: 19 (8)

EBUS indicates endobronchial ultrasound; ROSE, rapid on-site evaluation.

was 81%. For peripheral lesions, the diagnostic yield was 58% on the final cytology and 84% on histopathologic biopsy. While not statistically significant, the diagnostic yield for both cytology and histopathologic biopsy samples improved as the nodule size increased ($P=0.2$, Table 2). Most characteristics of the nodule and procedure did not affect the preliminary cytology, final cytology, or histopathologic biopsy diagnosis (Table 3). The only significant finding was that nodules diagnosed positive for neoplastic cells on ROSE and on the final cytology diagnosis were more likely to require only 1 to 3 passes for adequacy ($P<0.01$, Fig. 1). Otherwise, lesion size, lobe location, and segment location did not affect the preliminary or final cytology diagnosis. No association existed between the lesion's lobe or segment location and the number of passes obtained on ROSE.

In addition, the biopsy method (forceps vs. cryoprobe) showed that histopathologic tissue obtained by forceps biopsies was significantly more likely to have an atypical or nondiagnostic finding than a cryobiopsy ($P=0.04$). In addition, cryobiopsies had a higher percentage of biopsies that yielded diagnostic tissue (87%) compared with forceps biopsies (74%) (Table 3). There was no correlation between proceduralist or procedural volume by each proceduralist. Overall, 227 nodules were visualized on rEBUS examination (92%), and 21 were not (8%); however, there was no correlation between diagnosis and rEBUS visualization.

Diagnoses Concordance

Between the preliminary ROSE cytology and final cytology, 217 cases had concordant diagnoses, for an overall concordance rate of 79%. Similarly, between the final cytology and concurrent histopathologic biopsy, 217 cases had concordant diagnoses, for an overall concordance rate of 79%.

Three groups were analyzed based on ROSE results.

- (a) Positive ROSE results for malignancy: between the cytology diagnoses and histopathologic biopsy diagnoses, there was a high correlation between a positive preliminary ROSE cytology diagnosis and the final cytology diagnosis (99%), and between the preliminary ROSE cytology and the final histopathologic biopsy diagnosis (90%, Fig. 2). Of the positive ROSE/final cytology diagnoses but negative concurrent histopathologic biopsies, 2 had later histologic confirmation by histopathologic surgical resection, and 2 were clinically suspected to be malignant. One of these negative concurrent histopathologic biopsies had an intraoperative malfunction of the robotic platform. For the single false-positive preliminary cytology diagnosis (suspicious for malignancy), a concurrent histopathologic biopsy was negative for neoplasm. The patient underwent no follow-up histopathologic resection and

was followed clinically, with no radiographic progression of the nodules.

- (b) Negative ROSE results for malignancy: correlation between a negative ROSE preliminary or final cytology diagnosis and between a negative ROSE preliminary and a negative histopathologic biopsy diagnosis was not as robust (67% and 61%, respectively, Fig. 2). For false-negative cytology biopsies, 4 (14%) had concurrent histopathologic biopsies that were positive for malignancy.
- (c) Atypical/nondiagnostic ROSE: concordance for an atypical or nondiagnostic cytology diagnosis was the worst of the three categories (Fig. 2). Atypical/nondiagnostic ROSE preliminary cytology diagnoses often had a suspicious or positive biopsy diagnosis on the final cytology (67% of cases). For the concordance between atypical/nondiagnostic final cytology diagnoses, the distribution of final histopathologic biopsy diagnoses was relatively even, with 31% ending up as atypical/nondiagnostic, 36% as negative, and 31% as suspicious/positive.

Histopathologic and Clinical Follow-Up—False-Negative and False-Positive Rates

Forty-four nodules from 45 procedures (1 was a repeat ssRAB biopsy) did not have a concurrent histopathologic biopsy. Of those, 21 were given a reason, and 24 did not have a reason given. Poor visualization or loss of rEBUS signal ($n=6$) was the most common reason, followed by vessel obstruction ($n=5$), bleeding ($n=6$), and atelectasis/concern for pneumothorax ($n=4$).

Of the 44 nodules, 30 were followed clinically, and 14 had follow-up histopathologic specimens. For the nodules with histopathologic follow-up, the final cytology diagnoses were distributed as follows: 8 were positive for malignancy, 1 had atypical cells present, and 5 were nondiagnostic. Of those follow-up histopathologic specimens, 12 were positive for malignancy, and 2 showed benign findings (necrotic granuloma and fungal infection). All 8 nodules that were positive for malignancy on cytology were also positive for malignancy on later histopathology. For those clinically followed, the nodules were considered benign in 5 patients, malignant in 14, and unknown due to lack of follow-up in the remaining 10.

There were 5 false negatives in patients with negative cytology and negative concurrent histopathologic biopsy, resulting in a false-negative rate of 12%. Subsequent repeat histopathology biopsies were positive for malignancy. There was also 1 false-positive cytology case, resulting in a false-positive rate of 1%. In this case, the final cytology was suspicious for squamous cell carcinoma; however, the histopathologic biopsy revealed ulcerated bronchial

TABLE 2. Size Distribution and Diagnostic Yield

Lesion size (cm)	No. cytology specimens, (% of 250), n (%)	Final cytology, diagnostic yield (%)	No. Histopathological specimens, (% of 250), n (%)	Histopathological, biopsy diagnostic yield (%)
Overall	249 (100)	61	198	81
<1	75 (30)	54	61 (30)	77
1-2	98 (39)	64	79 (38)	79
2-3	51 (20)	60	43 (21)	81
3-4	7 (3)	86	6 (3)	83
>4	10 (4)	80	9 (4)	88

TABLE 3. Predictors of Diagnosis

Variables	Preliminary cytology diagnosis, <i>P</i>	Final cytology diagnosis, <i>P</i>	Histopathological biopsy diagnosis, <i>P</i>
Size	0.532	0.43	0.995
Location (lobes)	0.554	0.995	0.935
Location (B1-10)	0.501	0.89	0.76
Atelectasis risk of location	0.0978	0.513	0.761
Peripheral nodule	0.595	0.888	0.368
Physician	0.459	0.896	0.145
EBUS-visualization	0.724	0.899	0.478
Forceps vs. cryobiopsy	0.582	Not assessed	0.04
Passes	< 0.001	0.02	0.398

EBUS indicates endobronchial ultrasound.

mucosa with chronic inflammation and necrotic debris. The patient was followed clinically, with no progression of disease.

DISCUSSION

This study demonstrates high diagnostic concordance between a preliminary positive malignant result on ROSE, the final cytology diagnosis, and the histopathologic diagnosis. The correlation of diagnostic yield between ROSE results, final cytology, and histopathologic results was not as robust in nondefinitive and negative-for-malignancy ROSE cases. Lung nodule cryobiopsies had a higher diagnostic tissue yield (87%) than forceps biopsy (74%), with the latter providing more nondefinitive results (nondiagnostic/atypical diagnosis). A maximum of 3 FNA passes was shown to be the plateau for cytology sampling, and the likelihood that the nodule will have a nondefinitive result increases with each pass. Finally, nodule and operator characteristics did not affect the cytology or histopathologic diagnosis.

We observed a high rate of concordance between positive diagnosis of malignancy, preliminary ROSE, and the final cytology and histopathologic diagnoses. Notably, this finding was not associated with a difference in size, radial-EBUS confirmation, or operator. For other ROSE diagnoses, such as negative-for-malignancy or

nondiagnostic/atypical, a concurrent histologic biopsy is recommended, as the concordance rate is not as robust. We hypothesize that the reason for this discrepancy is that malignant cells tend to be more readily aspirated, whereas negative findings, such as organizing pneumonia, are more difficult to obtain through needle aspiration due to the presence of fibrous and more cohesive tissue. Other factors that could influence these findings include the heterogeneity of the tissue, vascularity of the lesion, and needle size, all of which were not recorded for this study. To our knowledge, this is the first study to evaluate the details of concordance by diagnostic category (positive, negative, and nondiagnostic/atypical), whereas prior studies have focused solely on overall concordance. A retrospective single-institution study of ssRAB biopsies for peripheral lung lesions found an overall concordance rate of 89% between the preliminary slide read from needle aspiration and the final pathology results. Concordance was 85% when needle aspiration results were compared with forceps biopsies, and 92% when compared with all specimen result.¹⁸ Granular details for concordance were not evaluated. Another retrospective single-institution study showed that the overall final diagnosis of their cytology samples was concordant in 51% of their diagnostic samples.¹⁹ The overall concordance rate of our study fell between those of the 2 prior reports, with 67% concordance between the preliminary cytology and the histopathologic biopsy and 79% between

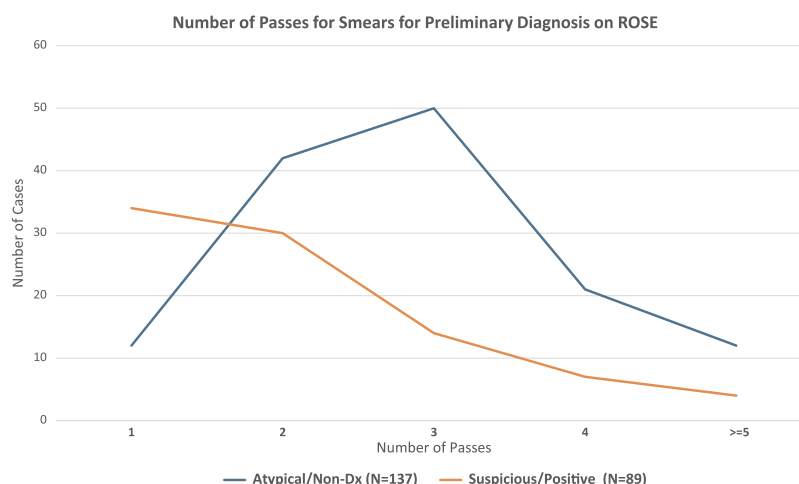


FIGURE 1. Number of passes for smears for preliminary diagnosis on rapid on-site evaluation. ROSE indicates rapid on-site evaluation; non-Dx, nondiagnostic result.

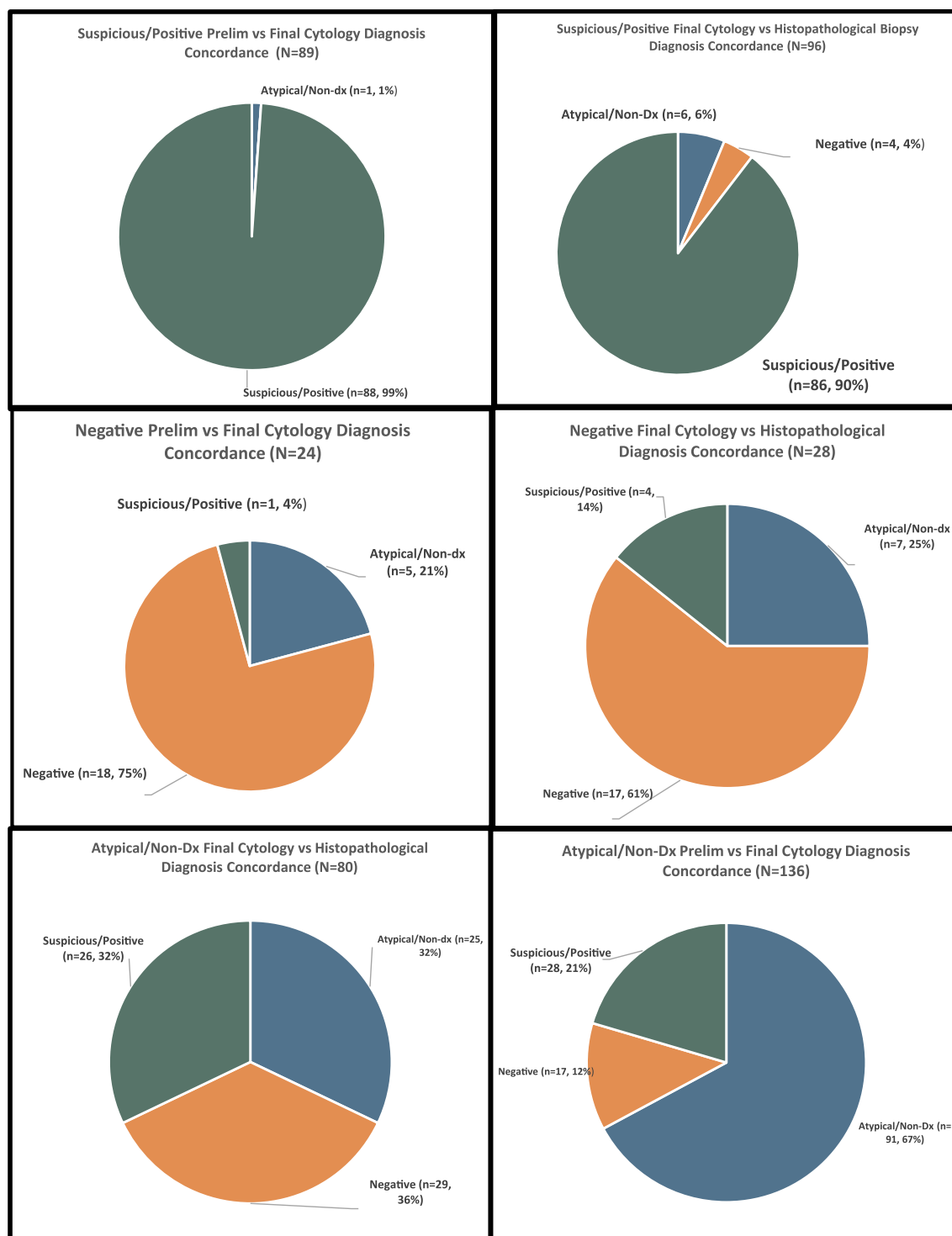


FIGURE 2. Concordance between positive, negative, and atypical preliminary cytology diagnoses versus final cytology diagnosis, and between positive final cytology diagnosis versus histopathologic diagnosis. Non-Dx indicates a nondiagnostic result.

the final cytology diagnosis and the histopathologic biopsy. Finally, our study's overall false-positive rate (1%) and false-negative rate (12%) are comparable to findings reported in other studies.^{19,20}

Cryobiopsies from lung nodules obtained by ssRAB had a higher diagnostic yield compared with forceps biopsies, with forceps biopsies more likely to result in atypical/nondiagnostic diagnoses compared with

cryobiopsies. Other studies comparing cryobiopsy with forceps biopsy did not find a significant difference in diagnosing malignancy between forceps and cryobiopsy; however, they found that cryobiopsy has a greater diagnostic yield.^{20–23} We have several potential explanations for the higher diagnostic yield from cryobiopsies compared with forceps biopsies. One is that the tip of the cryobiopsy probe does not need to be in the center of the lesion to obtain material. The probe can be positioned not only circumferentially to the lesion but also tangentially or en face. Forceps biopsy has the disadvantage of requiring an en-face approach to the lesion and is associated with a higher likelihood of crush artifact.²⁴ In addition, the cryobiopsy probe can biopsy fibrotic tissue; the distance from the probe to the lesion is more predictable compared with forceps, and the volume of tissue obtained tends to be larger.⁸

The optimal number of passes for ROSE diagnosis of lung nodules is 3 when using ssRAB, especially for nodules with a high suspicion of malignancy. Any additional passes for ROSE diagnosis lead to diminishing diagnostic results, adding unnecessary procedural time and radiation to the patient and team members. If additional passes are desired for diagnosis, we suggest dedicating them instead to cell block processing or switching to cryobiopsy to obtain diagnostic tissue. We emphasize that this benchmark of 3 passes is only for ROSE diagnosis, and additional passes should still be collected for any additional ancillary studies, including but not limited to molecular studies and PD-L1 testing, microbiology, and flow cytometry.

Understanding which types of nodules are best suited for ROSE and limiting the number of passes reduces the risk of unnecessary radiation exposure for both staff members from cytology and the operating room. Recent studies have found the mean total time for radiation exposure in the ssRAB procedure combined with cone beam CT is ~6 minutes, with an equivalent whole-body dose ranging from 0.37 to 1.48 μ Sv, depending on personnel.^{27,28} Additional passes are also more likely to be bloody and thus more difficult for interpretation by cytopathology personnel. An internal study to be published showed that ROSE added, on average, 21 minutes to the procedure, and there was no clinically meaningful difference in the diagnostic yield between cases performed with or without ROSE.²⁷ Additional studies have also supported our internal observations that there is no significant difference in the diagnostic yield or sensitivity for malignancy for cases that utilize ROSE and for those that do not.²⁸ Currently, our institutional approach is to use ROSE during ssRAB procedures in cases that include large pulmonary lesions (>3 cm) with heterogeneity on PET-CT or on CT scan, or if the target is the wall of a cavitory lesion.

While nodule and operator characteristics did not affect cytology or histopathologic diagnosis, there was a trend of higher diagnostic yield as the nodule size increased. There was no significant difference in the diagnostic distribution between peripheral lesions and central lesions. For peripheral lesions, the diagnostic yield for this study was 58% based on the final cytology diagnosis and 84% based on the histopathologic biopsy diagnosis. In the literature, there has been a wide variation in the reported diagnostic yield for peripheral bronchoscopy, ranging from 39% to 88%.^{17,29} For ssRAB cryobiopsies, a prior study reported a diagnostic yield of up to 97%, which is higher

than the 89% reported in this study. However, their definition of diagnostic yield is unclear in the methods. Our findings support another study, which also found that lesion size and location were independent of diagnostic yield, and the only statistically significant predictor of yield was the number of passes.³⁰

The limitations of this single-center retrospective study are multiple. Not all patients had confirmatory histopathologic biopsy or resection specimens, which may alter concordance rates. In addition, this study was not randomized and did not compare the diagnostic distribution and yield for nodules biopsied by ssRAB with those biopsied by conventional EBUS or CT-guided biopsy methods. In addition, it is worth noting that cytology on ROSE has certain limitations. The preliminary ROSE sample relies on staining of air-dried slides, which can introduce artifacts such as artificially enlarged nuclei,^{31,32} and the slides evaluated can have less tissue than the final cytology or histopathologic biopsy sample. Finally, as mentioned above, we did not note the heterogeneity or vascularity of the lesion, nor the needle size, which may have affected our results.

In conclusion, cytology and histopathologic biopsy samples obtained by ssRAB exhibit good concordance rates, particularly for nodules that are positive for malignancy. There are low false-positive and false-negative rates for both malignant and benign nodules. This study also demonstrates that only 3 passes for ROSE analysis are sufficient for adequacy in malignant nodules, which can improve procedure time and should be followed for tissue acquisition with cryobiopsy over forceps biopsy for optimal diagnostic tissue. While the diagnostic yield is not affected by the location of the nodule, it improves with larger lesions.

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